

Quick Check

Lesson 6: Where Does a Maize or Bean Plant Grow? Measuring Growth and Regions of Growth in Seedlings

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 6

A farmer in Nakuru plants maize and beans. A student predicts where a young root becomes longer. Which prediction best matches how growth in length happens in a root?

- A** Growth happens only at the part of the root closest to the seed
- B** Growth happens mainly in a region near the root tip, where cells divide and elongate
- C** Growth happens evenly along the whole length of the root
- D** Growth happens only in the oldest part of the root, far from the tip

Learners mark a young bean root with ink every 5 mm starting from the tip, then return it to moist soil. After 48 hours they measure the distances between the marks. Which marks will have moved furthest apart?

- A** All marks will have moved apart by the same amount
- B** The marks in the oldest part of the root, furthest from the tip
- C** None, because roots only grow in width
- D** The marks nearest the root tip

In the seedling investigation, why do learners mark the root at equal intervals (for example every 5 mm) from the tip?

- A** So that any change in the distance between marks shows which region has grown
- B** So that the ink dries faster on the root
- C** So that the root grows straighter
- D** So that the seedling grows faster

A maize plant was 4 cm tall on day 7 and 32 cm tall on day 21. What was its average growth rate between day 7 and day 21?

- A** 2 cm per day
- B** 1.5 cm per day
- C** 4 cm per day
- D** 28 cm per day

On an S-shaped growth curve of a maize plant (height against time), what does the steep middle part show?

- A** Growth has stopped because the plant is mature
- B** Growth is rapid, with active cell division and photosynthesis
- C** Growth is slowing because resources are limited
- D** The seed is still using stored food and growth is slow

A group is writing a Claim-Evidence-Reasoning statement about where a root grows. Which of these is a piece of EVIDENCE?

- A** Growth in length is concentrated at the root tip.
- B** All plants grow in the same way.
- C** This happens because cells in the meristem divide and then elongate.
- D** The marks nearest the root tip moved 8 mm apart in 48 hours, while marks higher up stayed the same distance apart.

Which statement best defines growth in a plant?

- A** Any change in the shape of the plant, with no change in size
- B** A temporary increase in size caused by a seed absorbing water
- C** An increase in height only
- D** A permanent increase in size and dry mass

Which plant can become thicker through secondary growth, and which tissue is responsible?

- A** Bean, through the vascular cambium
- B** Maize, through the vascular cambium
- C** Maize, through the apical meristem at the shoot tip
- D** Bean, through the apical meristem at the root tip

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).